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Original Article

Remote UX Methods for Kansei Engineering – A Case-study in Assessing Emotional User Response to Computer Mice through Remote Interviews and Online Data Gathering

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Abstract: During the pandemic, UX methods employed in hardware design were restricted or postponed. The typical controlled UX studies were out of reach, as access to users and their exposure to products was restricted. This offered an opportunity in the design of computer mice, found to be in higher demand for working from home. This work describes the development of an adapted method for gathering UX, driven by Kansei Engineering (KE), properly supported by a purposely designed online platform to guide, provide tasks and survey users, KESo. The hypothesis is that remote UX methods allow collecting user feedback with no major loss of subjective data providing similar insights as traditional methods. The emerging idea is that these methods can and will remain useful to support product design. Remote UX can also be autonomous, reducing resources required, thus promising scaling participation, and extending capabilities of gathering emotional responses by adopting new technologies.

Keywords: User experience methods, Remote UX, Online UX, Computer mice

1. Introduction

In product development processes supported by User Experience (UX) studies, emerging data often depends on the degree of interaction and engagement between participants, interviewers and overall accessibility to products. More detailed feedback and potentially valuable insights are expected to be retrieved when a higher level of exposition in Product Experience (PUX) is possible. Aside from the UX method employed and the design of the experience itself, this usually means direct access to physical products by the users and a longer experience in general, surpassing mere initial exposition and first impressions. However, in the process of designing a UX study, a preemptive assumption by both researchers and industrial professionals is the requirement of using live interviews, where both users and interviewers engage in person and share, even if for partial moments of the study, the same space. To some extent, and beyond survey methods typically aimed at capturing data from larger demographics, a common assumption is that UX interviews need to be done in person. The present study aims at challenging such perspective by reporting a UX study, forcedly performed remotely, adapting it from its initial design to cope with the global pandemic.

1.1 A proposed model of Kansei Engineering in an applied context

Kansei Engineering (KE) sees the Kansei concept applied to product design. Consumer's feelings and overall impressions are regarded as perceptual design elements and, along with the product or service physical attributes, through a reductionist approach, are broken down and prioritized. Product development processes make use of known design factors which affect product performance. In a case within computer peripherals, specifically computer mice, the main keys clicking feedback is regarded as the goal of an engineering design, effectively changing both the haptic and acoustic cues afforded to the users. To better understand the consequential UX of such changes, a complete KE-based UX method was envisioned to properly assess a set of prototypes within target user populations. With such case study in-hand, researchers

emerge with two very distinct data sets: a quantified and experimentally measured mouse click feedback; and a very subjective emotional response from users to such feedback. Here, statistical analysis methods, ranging from descriptive statistics to machine learning methods, are available to KE researchers and professions to establish and identify relationships between measured performance and subjective user feedback. Depending on the chosen and combined analysis methods, results can range from overall holistic experiences, such as rankings and ratings of preference, to mathematical relationships between design factors and specific user emotions. Outgoing from this model, summarized in Figure 1, the authors derived a Kansei UX model. While applied to computer mice clicking, where haptics and acoustic metrics are the measurable product performance, the model can be regarded as generic and applicable by KE researchers and professionals, when seeking to enclose and bridge both data emerging from both design and engineering as UX studies.

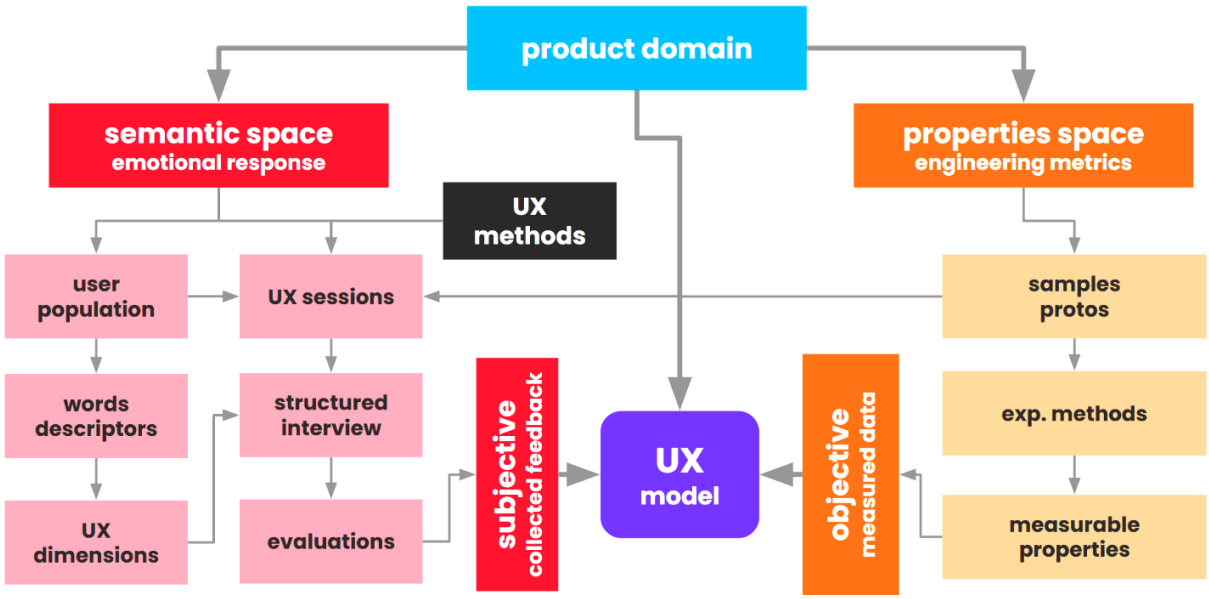


Figure 1: A high-level Kansei Engineering methodology, applied to a product development process. The flow of activities is chronological in a top-down approach, albeit some can happen in parallel. The end goal resides in achieving a predictive UX model that, if valid, allows preemptively defining product emotional responses through engineering specifications. In the current case, design factors are well known, and product performance is measurable, hence quantifiable; while a comprehensive KE-based UX study gathers and captures the user experience and verbal feedback, establishing a set of subjective data. Statistical analysis methods are then applied to bridge both sets of data, identify mathematical relationships between measured performance and user emotional response.

1.2 Aim and research questions

This study and its research questions (RQ) made use of the UX of selected characteristics of computer mice, specifically the haptic and the acoustic cues emerging from mice clicking. Prototypes were engineered to afford users different levels of such cues, as well as sensorial cross-modal effects. Meaning, prototypes offered well distinguishable and distinct mice clicks. Not only was the quantification of overall user emotional response through KE methods implemented, as well any impact of performance was assessed due to changes in particular feedback cues.

However, in the design of such an experience, and while applying the first steps of KE methodologies, the researchers saw their access to their most valuable resource, the users, impeded by the global pandemic. Specifically, for an unforeseeable amount of time, no live interviews could occur and exposition to the prototypes could not be simply handed over. A redesign and adaptation to the original methods envisioned was seen necessary to cope with the new conditions. Inherently, the researchers were now facing a new dilemma, one that constitutes a new and the two main emerging research questions:

RQ1: How can UX methods be developed and adopted to perform remotely to cope with any research scenarios with limited / restricted access users?

RQ2: What is the yield, validity and value of the gathered subjective data taken from remote UX testing?

The aim of this paper is, thus, to propose an open framework on performing remote UX data collection, within a Kansei Engineering methodology, supported by a case-study using computer mice as an application.

1.3 Case study, its relevance and context

Product choice for this research benefited from the development process of a new series of computer peripheral products, specifically a high-end office and productivity mouse. Such common electronic devices, familiar to any using a computer, can be easily found both at the workplace as well as at homes. Familiarity with such devices lessens the period of adaptation to the case-study, even if new samples are provided to users.

However, a challenge emerges from introducing changes to common experiences, particularly when long-term branding has been associated with features that dominate Product Experience (PUX). In this case, computer mice main keys feedback, also known as the mouse click. Surely, both design and technology evolved, but the main mouse click feedback has remained mostly coherent throughout many product generations, with minor, if even perceivable, changes due to component integration and system design. Computer mice and specifically the mouse click, were thus selected as a case-study in researching remote UX collection strategies. Here, the researchers chose to assess the overall level of intensity for the cross-modal effects between haptic and acoustic cues. In such exploratory work, focusing on a few selected product properties instead of a holistic product experience, is expected to narrow down the feedback given by the user, aiming at better correlations to measurable product properties.

Interestingly, the context imposed during the pandemic, including Work from Home (WFH), led people to quickly adopt remote working methods, providing an adapting and evolving context where computer mice were a main work tool. This study was then exploring new market trends that could have emerged in this new global scenario.

2. Literature Review

2.1 Kansei Engineering

Kansei Engineering (KE) is a product development methodology that integrates the emotional responses of users into the design process. Originating in Japan, it aims to capture and quantify the sensory and psychological impressions that products evoke in consumers. By systematically linking these subjective experiences with specific design elements, Kansei Engineering ensures that the final product not only meets functional requirements but also resonates emotionally with users. This approach is particularly valuable in enhancing user satisfaction and creating products that evoke positive emotional responses, ultimately leading to a stronger market appeal (Bongard-Blanchy et al., 2013; Choi & Jun, 2007). KE methodology is specialized in translation users' affective values into concrete product design parameters (Schütte, 2005). The comprehensive procedure for a KE methodology approach can be found in the literature (Nagamachi, 2011; Schütte, 2006; Schütte et al., 2004) and for which, a summarized conceptual model is shown in Figure 2.

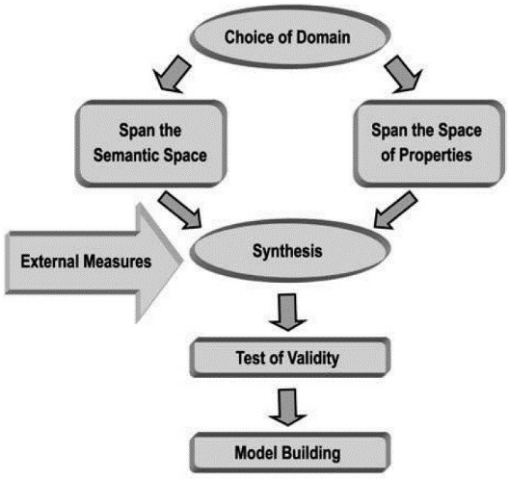


Figure 2: Model on Kansei Engineering method

In short: A group of products (domain) is described using affective semantic terms (Semantic Space) and using physical technical terms (Space of Properties). In a synthesis step these two vector spaces are interlinked using different types of algorithms (Marco-Almagro & Tort-Martorell, 2012; Schütte, 2005; Schütte et al., 2004). This model can then be used to optimize a final product solution.

2.2 User experience panel composition

In any approach made by a company in sensory analyses, a panel of human assessors composed by testers, judgers, and trainers is required (Mauter, 2001; Mauter & Katzki, 2003; Mittermeier et al., 2010; Spence & Zampini, 2006; Yantis, 1992). Source of UX information can occur at all levels of PUX, from naïve customers to expert teams (Falk et al., 2014). While naïve populations can provide insights closer to their own individual needs not yet known to product developers, well-trained panelists allow sensorial consistency, one which defines the product and, ultimately, the brand (Hagtvedt & Patrick, 2009; Hassenzahl, 2008; Lindstrom, 2005; Schmitt, 1999). Several factors must be considered to both select and form a user panel (Meilgaard et al., 1999, Chapter 9) . Health related issues, specifically hearing and haptic sense ability must be regarded as implicit and limitative in selection.

The use of two panels is common in Sensory Analysis methods, hence experiments performed in creating lists of attributes with distinct experts and naïve teams have led to comparisons between the effectiveness of using both groups (Giboreau et al., 2001; Poirson et al., 2010; Reisinger et al., 2006; Soufflet et al., 2004). These researchers found consistent results but also highlighted differences not only in the feedback but also the ability to verbalize perceptions of specific sensorial cues. While experts exhibited a higher ability in semantic descriptive information, Reisinger et al. (2006), found exceptions in the naïve population used, “those with a lot of experience.”

It is an assumption in this work that in the market of computer mice, there are many exceptions in a naïve population: those that, albeit unknowledgeable to the particularities of haptics and acoustics, use the device intensively in their daily routine. In this context, the researchers, beyond making use of an expert panel, also used a naïve panel. Yet, the latter is considered composed of target users for computer mice and were selected within the context of a medium-to-high level of computer proficiency.

2.3 Remote UX methods

Remote data gathering, as in the examples of census, marketing and other activities, are common practices and are usually achieved through surveys (post or online), phone calls and other technologically backed communication methods.

Prompted by the pandemic, the remote UX methods herein reported differ from the previous both in scope and goals. These methods retain the similar aims of conventional UX interviews to support design processes through product exposure and capturing user feedback but without traditionally co-locating all participants and products in the same space. Here, Remote UX methods is, thus, a structured process of gathering user experience where the connection between the interviewers, those hosting the study, and the interviewees, the target user population for PUX, is achieved by digital and online/internet-based means. The moments of gathering data can encompass from workshops and group sessions to single-user interviews by use of video conferencing, but also take the form of email exchanges, surveys, online platforms and shared teamwork tools, e.g. sharing digital whiteboards; while complemented and supported by written, video, animated or any digital form of tutorials and guidelines. Product experience is promoted with the user by providing shipped products, realistic PUX activities, games, and tasks which can be easily and readily accessible online, hosted by the study’s platform or third-party sites. Complementing these specific PUX scenarios, including free users’ own individual contexts, e.g. Working from Home (WFH) can be remotely provided. The remote UX methods aim at complementing traditional UX methods or replacing them in their entirety, according to the scope of the study.

Remote UX methods adopt any form of data gathering in UX studies (Grunwald, 2008, Chapter 37), taking subjective answers from recorded or written open answers and verbatim collection but also quantifiable objective ratings, rankings and scores given to products, specific product features or experiences, including emotional responses. In the latter, affective engineering methods, in particular a full KE Quantification Type I methodology (Nagamachi, 2011; Schütte et al., 2004) is employed throughout this study.

2.4 Previous research employing remote UX methods

In the context of sensorial analysis, the automotive industry provides many research cases and studies regarding haptics and hearing. Published research examples are available for sensorial evaluations of leather and other textile surfaces (Falk et al., 2014; Giboreau et al., 2001), automotive HMIs (Gaspar, 2013; Gaspar et al., 2016; Pitts et al., 2009, 2012; Wellings, 2008; Wellings et al., 2010), engine sounds (Altinsoy, 2013; Nosulenko, 2013; Poirson et al., 2010), tapping dashboard sounds (Montignies et al., 2010) and closing door sounds (Jo et al., 2014; Parizet et al., 2008) and power window sounds (Volandri et al., 2018). Yet, no form of remote UX methods was found in this review.

While remote data gathering, through the form of internet-based surveys is a common, well-known and widespread procedure, it does not necessarily tackle sensorial UX or product experience overall as in the present case-study. A few examples found in the literature and reference studies are herein listed. Takada et al. (2019) adopted a methodology of exposing users to prerecorded sounds from closing automotive doors, in order to evaluate overall quality of car build from acoustic metrics. This methodology allowed researchers to efficiently expose a larger panel, 40 participants, to isolate acoustic feedback, while avoiding the need to expose them to multiple cars. Used audio samples were afforded equally to all participants, making the stimuli presentation well controlled and removing possible variations from generating feedback experimentally for each user. More importantly, this sensorial modality isolation method removed the quality bias possible to introduce by knowing which car, i.e. model and brand, was the source of each audio sample. San Diego and McAndrew (2009) focused their research primarily on UX data collection using eye-tracking equipment. However, a second- smaller- study is reported on a remote observational study of online learning activities. Eika et al. (2019) found reference data on hand grip and lower limb strength using the Nintendo Wii balance board by remotely assessing users experience. Walsh et al. (2010) aimed at identifying cultural differences on the experience of mobile phones, where parts of the data collection was performed online.

While references might be scarce, they serve as a base for future explorations. Although unmentioned, methods in UX as those provided by Grunwald (2008, Chapter 37) and Hartson and Pyla (2019, Chapters 24, 25) can surely be used or adapted for remote UX and online data collection.

3. Study Procedure and Materials

The case study was carried out in three subsequent steps following the general procedure presented earlier in Figure 1 and Figure 2. Figure 3 showcases the concrete steps. Firstly, a benchmark was carried out leading to the creation of the Semantic Space. The second step featured the collection and prioritization of emotional features related to haptic and auditive feedback from computer mice switches. Both of those steps required collection of UX data, which was partly done in person prior to the pandemic and remotely online during the pandemic. In a final step the two spaces were interlinked together and relations between the affective and physical features established.

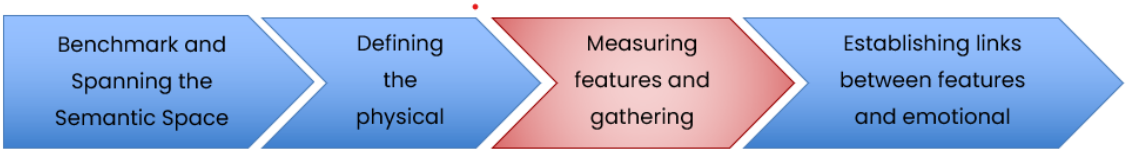


Figure 3: Procedure applied for the case study

3.1 Defining physical emotional features - materials

As functional samples for this study, 2 models of commercially available computer mice, Logitech's MxMaster3 (MM3) or MxAnywhere2 (MA2), as seen in Figure 4, were provided, with 4 variations for each model. These mice were internally modified, introducing different types of electromechanical switches for their main keys' actuation. Selection of switches was such that they did not differ substantially in haptic parameters, specifically operational force and actuation travel, yet afforded well noticeably different acoustic feedback. The choice of switches was made through benchmarking available switches from the market and, by use of an internal expert team at Logitech, paired with an internal UX test (N=20). In the latter, the switches were categorized into different noticeable levels of (perceived) loudness and acoustic sharpness.



Figure 4: Commercially available products from which the prototype samples were derived left: Logitech MX Master 3; right: Logitech MX Anywhere 2.

Apart from the 2 different available mice models, all samples were visually indistinguishable, and identification was only possible through abstract labels known only to the researchers. Labels were defined according to model-switch-(left/right), and different switch types labeled A, K, O, P. These labels will be used throughout the following figures and tables.

3.2 Feedback from computer mice clicking

With a computer mouse, the user physically manipulates the peripheral over a surface, e.g. the desktop, and displaces an on-screen pointer towards an object of interest, such as a menu option or a target. This is an exploratory haptic action, where the detection of the object of interest is achieved through visual cues: pointer spatial location over the object and change of the object aspect. Upon achieving and identifying the object location, the intent of the user acting upon it is achieved through a finger press. This action is, by itself, a separate psychomotor event composed of intentionally displacing the finger downwards, pressing the mouse keyplate. Mouse clicking is, thus, a consequence of active touch, involving both tactile and kinesthetic perceptual processes, also known as haptics. The mouse, in return and due to its technological implementation, affords immediate and progressive sensorial feedback: a sense of displacement and increase in force. This promotes an identification of positive communication and interaction between Human and Machine, linking intention and action for the task to perform. Besides this feedback, and throughout the interaction, a distinct and unique event occurs, a sudden variation in force, resultant of the mechanical response embedded in the device. This affords the user a distinguishable confirmatory haptic cue: the instant where the user's intent is achieved and is sensed from the device. Besides the force drop, resultant mechanical vibrations, structurally transmitted to the user's finger, and airborne acoustics are perceivable, thus enriching and reinforcing the confirmatory cue. Implicitly with this event is the sense of usability of the device. The "click" feedback is, thus, a complex perceptual summation of both haptic, tactile and acoustic cues. From the user perspective, integrating these multi-modality cues results in a confirmatory mental process, a sense of achievement for the desired interaction. Embedded in the feedback, there are recognizable emotional qualities emerging from the experience. This elicits a subjective perception of the quality of the interaction and, at a higher level, a contribution to the holistic perception of product quality and promotion of brand sense. On the opposite perceptual perspective, the lack of adequate feedback can result in very strong emotional negative responses, such as frustration and a sense of inability to perform the task at hand. The click feedback is, implicitly, a great contributor to the sense of overall product quality and its UX, capable of defining a company's brand sense.

The measurable mechanical behavior of a mouse click can be characterized with force-displacement curves. These are called 'haptic profile' (Ayas & Eklund, 2010; Feit et al., 2016; Nagamachi, 2011, Chapter 10; Weir et al., 2004) or 'haptic feedback' (Rydström, 2007; Sokhanvar et al., 2012). Analyzing engineering parameters taken from haptic feedback measured data is fundamental to predict how the interaction afforded and, consequently, product quality will be perceived and distinguished from others. A literature review on haptic profile research, including previously identified metrics for these profiles, while extending this exploration to a comprehensive array of metrics can be found in Valverde et. al. (2019). Similarly, research on the acoustic emissions typical of push-buttons, regarded as impulsive sounds, can be found in Valverde et. al. (2018). The taxonomy of haptic interactions can be found in Hatzfeld & Kern (2009, Chapter 1) and further definitions for confirmatory haptics, passive haptics and active haptics definitions can be found in (Breitschaft et al., 2019).

3.3 Experimental data in haptics and acoustics

Mice samples and both left and right keys had their haptic and acoustic feedback experimentally measured and characterized, using methods similar to previous research in the field (Valverde et al., 2018, 2019), albeit adapted to the case-study and internal capabilities and experimental procedures. A total of 16 haptic metrics and 5 acoustic psychoacoustic metrics were considered for further data analysis. Some of these are typical industry standards, e.g. operational force and A-weighted instantaneous sound pressure level, while others are internally explored at Logitech to further understand UX relevance and mouse performance. From observing the data, the original assumptions were further confirmed: samples available had similar haptics such as the required force to click but noticeably distinct acoustics. A notable exception in haptics was a combination of metrics related to the instant of the click itself which differed sufficiently in two samples (same switch, different mice) to be perceived and rated differently by users for some of the UX descriptors.

4. Kansei Engineering and Data Collection

4.1 Benchmark and spanning the semantic space

A collection of 60 different computer mice was gathered, with products spanning from different brands, mouse types, different portfolio eras (oldest mouse available from 1990s) and even distinct clicking technologies. Additionally, other mice were brought into the study by each of the users participating in this stage.

The goal for this stage was proposed in eliciting semantic space on, specifically, mice clicking interactions. Hence, the researchers focused the attention of their panels collecting expressions emerging solely as haptic and acoustic cues and feedback from mice primary function: mains keys. Users were also asked to ignore and disregard other experiences taken from available samples, such as mice size and shape, secondary features, visual cues (i.e. colors), etc. At this stage, the semantic space should solely address mice clicking UX.

Logitech: the panel was composed of 19 Logitech employees in mice design and engineering departments. The session, pictured in Figure 5, introduced the goals of this workshop to the panel, via an overview presentation on KE methodology and the definition of UX descriptors / dimensions. Users would then freely explore all the available mice samples, through a period of 30 minutes, while writing down in post-its the individual verbal expressions emerging from each user's own elicited experience with the samples. Throughout the session, the researcher would encourage participants to swap samples frequently.

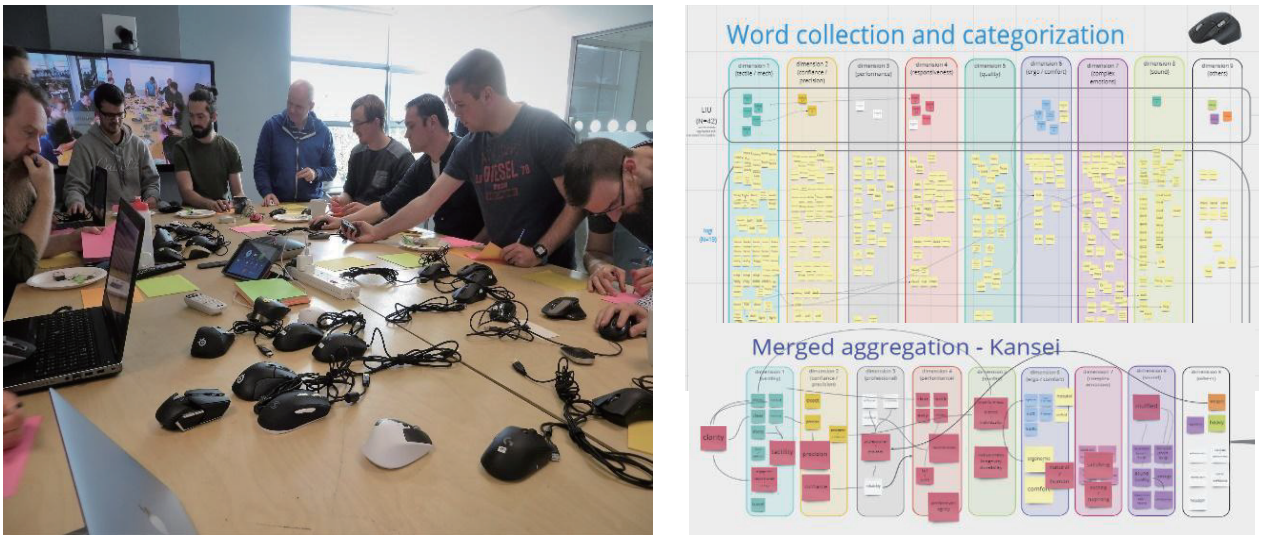


Figure 5: Left: joint session at Logitech; right: digital whiteboard from the already categorized collected adjectives generated during both Logitech's and LiU's sessions

LiU: users would annotate the expressions in virtual post-its (on a virtual whiteboard). Communication was ensured using an online conferencing tool. Before actual work with the data collection could start the users were introduced to those tools (Microsoft Teams and Miro). The data collection itself followed the same procedure as done in Logitech. Users brainstormed verbal expressions on the product and put them onto post it. After saturation was achieved, users worked in groups to cluster the words/post-its. These clusters then build the foundation of further work. In total 42 experienced users of computer mice participated in the LiU data collection.

Hence, the generation of the semantic space through exposition of benchmarked mice had a total of 61 participants, of which 18 were female, ages ranging from 21 to 63 years old, and declaring different levels of proficiency regarding computer mice.

With words and adjectives given through collected verbose and/or written annotations, a total exceeding 740 descriptors were collected. From these, some were identified as not being specifically related to mouse click feedback, hence removed from the pool. Additionally, 39 different descriptors were added from previous research and internal documentation. A final list of 690 descriptors associated with this study scope generated the user-driven semantic space for mice clicking UX. As shown in Figure 5, these were then digitally captured with a whiteboard tool, aggregate and categorized according to its scope and meaning.

Between the pre-pandemic presential workshop and the remote workshop held at LiU's, it was possible to observe notable different results, including in the collaboration between all participants and dynamics involved.

For the live workshop, a considerable amount of time and effort was required to find, book and allocate the required space as well as setting up the environment and materials, including sourcing real mice products. Participants were booked through email and calendar. From the start of the session, upon a presentation clarifying the goals and activities, motivation and curiosity led to an initial enthusiasm from the group. During the session, a permanent discussion and a collaboration between all participants could also be observed. The use of post-its as well as experiencing real mice samples, as well as trading products among participants, appear to have contributed very positively to the dynamics and mood of the workshop. This single session provided a large set of words in less than 2 hours, proving also efficient.

For the remote session, participants were invited through email and calendar to virtual meeting sessions. The total number of participants were distributed into 4 groups, each with its own session. The use of product images, from the internet, was efficient in communicating the scope of the workshop and allowed the researchers to collect and distribute them immediately without the need of sourcing products. The presentation, similar to the live workshop one, was given through videoconference. An open digital whiteboard was made available to all participants to write, edit and manipulate digital post-its or any other content they wished to add. The interaction between participants was conducted by videoconferencing etiquette, meaning only one participant would speak at the time. This may have limited spontaneity and engagement from users. Moreover, without parallel conversations occurring, interactions were also very limited by time. However, the simultaneous use (and novelty for some) of the digital whiteboard seemed very beneficial, allowing some observable virtual interactions like moving and even grouping their own and other post-its. To generate a similar number of words, more participants and total collection time, through multiple sessions, was required.

4.2 Defining emotional response features - descriptors (semantic space)

From the total number of adjectives elicited, a list of descriptors was aggregated and categorized by affinity, associating its meaning and looking for synonyms within given descriptors. A total of 9 dimensions/categories were identified and are shown in Figure 6. The same data in more detail can be seen in Figure 7, and applying a Pareto Analysis on the frequency

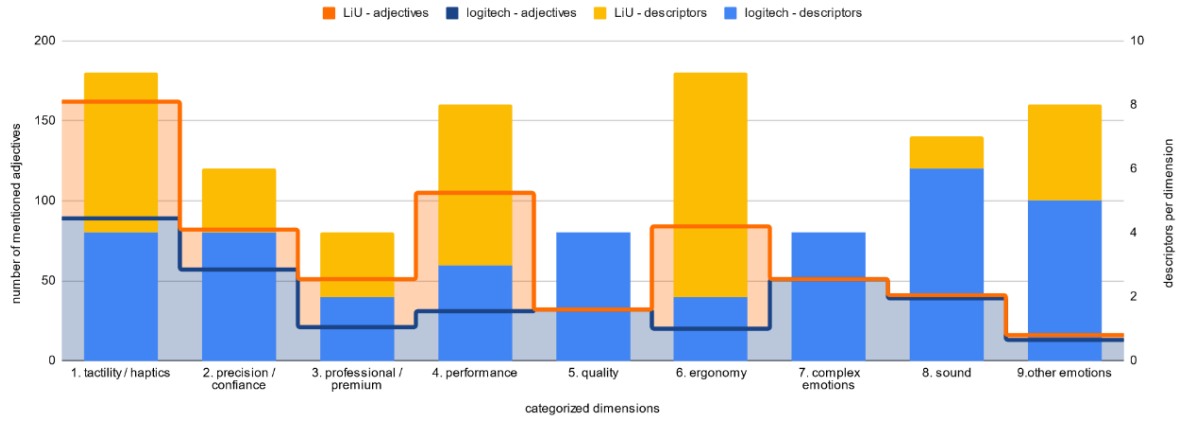


Figure 6: Summation of both Logitech's and LiU's number of total elicited adjectives, including repetitions (left scale) and unique given descriptors (right scale) aggregated by each UX dimension

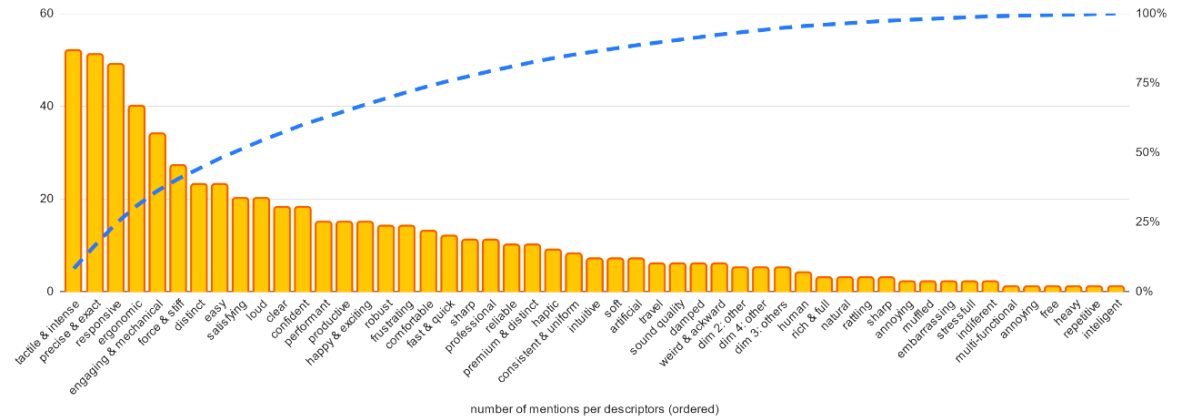


Figure 7: Pareto analysis of the number of total adjectives elicited and their frequency

of the adjectives used, it is possible to observe that the first 3 descriptors address directly the tactile feedback from the mouse click, as well as half of the total words given used fall into the first 7 descriptors.

From the total collection of KE words, a reduction to 15 affective descriptors was made through joint meetings within the expert panel. These words were expected to encompass most of the experience afforded from mice clicking. Final 15 descriptors: *loud, engaging, muffled, durable, professional, surprising, satisfying, responsive, distinct, robust, exciting, perceptible, damped, precise, tactile*.

4.3 Gathering emotional response

The initial plan for gathering data for our mouse click study involved inviting users to our labs for conventional UX data collection. However, during the preparation phase, the pandemic shut down all possibilities for in-person meetings with study participants. As a result, we developed a method to collect data remotely. Looking for previous references, few existing studies exist on remote UX data collection and were hardly adaptable to this case scenario. Given the circumstances, we devised our own method for remote data gathering.

The data collection software tool used was based on Kansei Engineering Software (KESo), developed by researchers at Linköping University to aid Kansei Engineering studies. KESo can conduct initial data evaluation, such as linking the Semantic Space to the Space of Properties, allowing us to understand how different properties influence participants' perceptions. This understanding enables us to design the intended perception by modifying properties accordingly. The KESo software was adapted and integrated with a new software tool developed by Logitech, allowing participants to perform various click tests and rate their experiences within the same window.

As shown in Figure 8, rooms at LiU's lab and Logitech's office spaces were allocated and equipped with eight prototype mice and a data collection computer. Participants were then invited to book time slots in these rooms to conduct the data collection independently. Each participant was required to perform several activities on a laptop made available, while using all the different mice. The interview was performed via video conference, where the interviewer was portrayed on one screen; while the KESo platform, including the randomly picked sample and providing the semantic differential scales, was shown on the second screen.

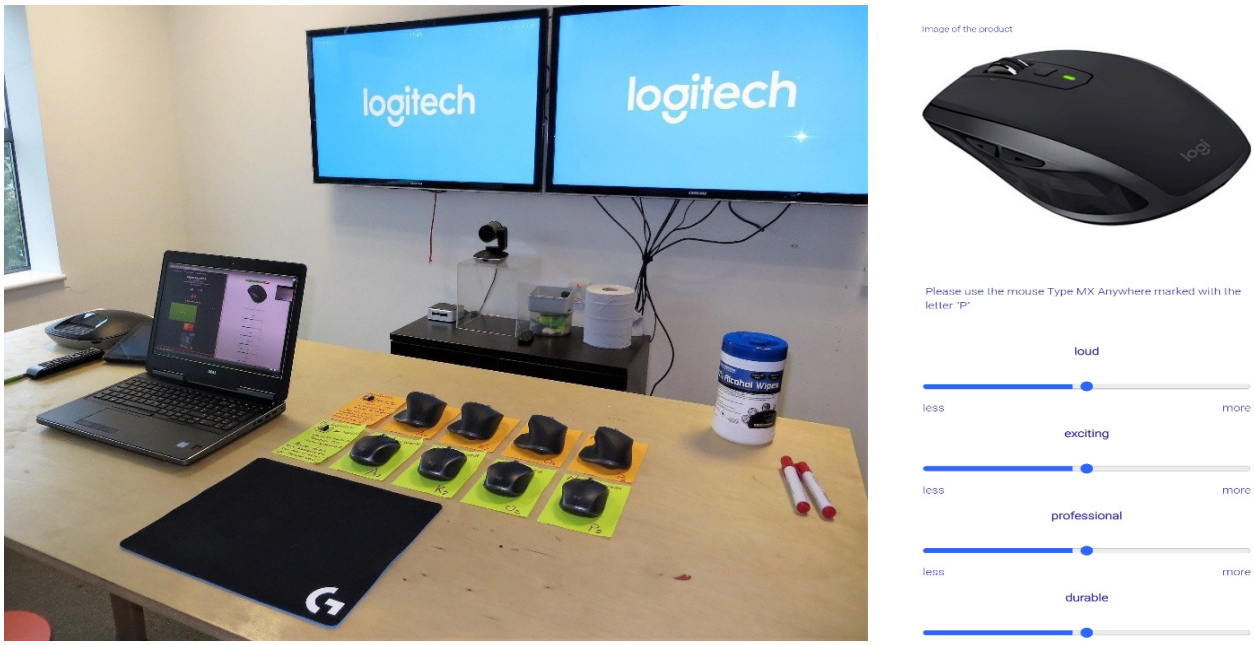


Figure 8: Left: Interview space replicating an office scenario made available at Logitech, providing a remote interviewing setting where samples were directly available to the user as well as the tasks and activities provided; right: partially screencapture of the KESo identifying the randomly picked sample toi use as well as the UX scales provided to the interviewee

To further extend the accessibility to the UX test, in a scenario where the participants could not be present in the test room, a fully remote setup was made available. In this case, the participants would either collect the mouse samples or receive them, e.g. at home, and perform the UX autonomously at any time of their preference. The KESo platform, via

webpage, was available to use remotely and researchers prerecorded a tutorial guideline video including the interview and rating process. This is available from the QR code in Figure 9.

Figure 9: Online Data collection guideline, link can be also visited at https://youtu.be/gr-poSLSfkQ.

5. Findings

The purpose of this section is just to provide some exemplary findings on the results. The results are internal to the researchers and to the institutions involved and are regarded as proprietary. The focus of this section remains on the findings on the UX methods and the contributions from this work within the framework provided.

5.1 Statistical analysis

To the measured experimental data as well as the quantified user experience, which averaged results are illustrated by Figure 10, a common but comprehensive analysis with a KE quantification type¬-I method was used to assert and identify relationships between data. The methods employed are herein listed albeit results not fully reported, just summarized: general statistical analysis with descriptive statistics, Factor Analysis (FA), Principal Component Analysis (PCA) including extraction with Varimax Rotation and Kaiser normalization, Pearson's Correlation, Analysis of Variance (ANOVA) including Tukey post-hoc tests, Main-effect analysis and interaction plots. Analyses were performed with a 95% confidence interval.

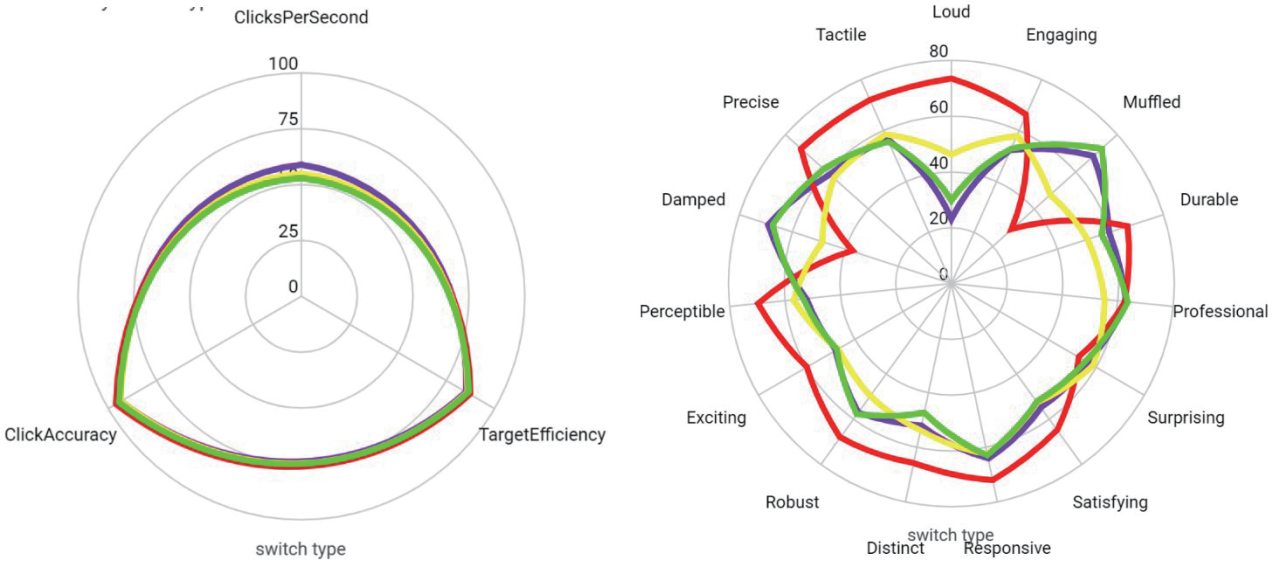


Figure 10: Left: measured task performance by users in 3 interview activities; right: radar chart means values of ratings given for each descriptor. Color bands identify different product samples adopted in this study and how they distribute across the UX ratings and measured performance.

From the descriptive analysis, mice were shown to be consistent in their experimental data for each key and sample. From the quantified user experience, provided tasks showed no performance impact from samples while their afforded UX was clearly rated distinctively.

Between measured metrics and UX descriptors, significantly strong correlations were found, highlighting that user given descriptions matched engineering metrics, e.g. descriptor loud was strongly correlated with measured A-weighted instantaneous Sound Pressure Level, $R(6) = 0{,}589$, $p<0.001$. Yet, non-significant correlations showed that some descriptors were not explained by the measured metrics. Additionally, several correlations were also found among the majority of UX descriptors themselves, hinting confounding effects: from the 15 chosen descriptors, many were likely to address the same experience dimension.

FA and PCA methods confirmed this observation. One single component of the UX data explained 46% of data variance, while the first five components explained over 80% of data variance. From the PCA plots, 3 distinct UX components were observable. These were interpreted as: 1. click quality, 2. click acoustics and 3. surprise/novelty. Further PCA with Varimax Rotation extends this understanding to 4 dimensions: 1. Quality and success of interactions, 2. Professional and premiumness of the click; 3. Click acoustics and 4. Novelty and surprise. While the former three were expected, the latter, a complex emotional response, was unexpected to the researchers. Further interpretation clarified that the users were surprised (and perhaps positively excited) by mice affording a different click sound, specifically quieter and less acoustically sharp than the traditional mouse.

ANOVA with Tukey post-hoc tests allowed aggregating the samples from their experimental data and, when correlated with the UX data, categorizing them according to descriptors. For example, for the descriptor loud, there was a significant effect of switch type to determine the given perceived loudness, $F(3, 148) = 142.59$, $p<.01$. Interpreting results, samples afforded to users were distinctively perceived as quiet, slightly quiet and loud. Similarly, for the descriptor precise, significant effects were observed, $F(3, 148)=3.35$, $p=0.021$, hinting that the perceived feedback, in this particular emotional response, was different among prototypes. Other descriptors, however, showed no distinction afforded by switch type. For example, descriptor professional showed no significant effect from the prototype variations, $F(3, 148) = 0.96$, $p=.414$, hinting that there was no detrimental effect for the perceived quality of the product in this category.

Interaction plots were also used to understand the effect of both mouse model and switch type on the UX descriptors. For example, both mouse model and switch had no interactions found for the descriptor loud, while a strong interaction was found for mouse model for the descriptor professional. In summary, Table 1 lists the significant effect of switch type at a $p<.05$ level for each descriptor.

Table 1: Results of the ANOVA with Tukey post-hoc tests for how significant ($p<.05$) the switch type was to determine users' answers on each descriptor

descriptor	by switch	descriptor	by switch
loud	yes	distinct	yes
engaging	yes	robust	yes
muffled	yes	exciting	yes
durable	yes	perceptible	yes
professional	no	damped	yes
surprising	no	precise	yes
satisfying	no	tactile	yes
responsive	no		

Main effect plots were also employed to better understand the experimental measured data and each of rated descriptors. This would further provide insights into the optimal engineering specifications for the click experience towards meeting specific UX dimensions.

5.2 Findings about the remote method

This section presents the findings related to the remote method employed in our study, linking them to the overarching research goals and aims. The investigation focused on spanning the semantic space of user experiences with computer mice, utilizing both live and remote workshops to gather data.

5.2.1 Exposition to mouse samples

During the live workshop, participants had access to a considerable array of mouse samples, while in the remote workshops, users were tasked with sourcing their own samples and exposed solely to product images. This difference in sample exposure is significant; familiarity with computer mice appeared to facilitate the generation of descriptive words

across both methods. Given that the product is commonly used by all participants, the familiarity has likely contributed to a more robust vocabulary. However, the diverse exposure to samples in the live workshop may have enriched participants' experiences, leading to a more focused generation of words, particularly concerning the perception of feedback quality.

5.2.2 Live workshop vs. remote workshops

The total collection time for the live workshop was approximately one hour. However, it is important to note that preparation for the workshop, including setting up the room and digitizing the produced content, required several additional hours. Both digital murals and written posts were found to be equally effective for data collection. Nonetheless, some participants using digital murals may have needed to familiarize themselves with the tool during the data collection process, which could have limited their contributions. To mitigate this issue, we recommend implementing a pre-workshop exercise that allows users to learn the necessary skills through a tutorial or simple game.

Analysis of the data revealed that both panels generated a similar magnitude of words describing mouse clicks, which could be categorized into a comparable number of categories. Yet, the remote session required over twice the participants and more collection time as different session were held. Still for the outcome, no significant differences in the volume of contributions were observed between the panels. If the product is well-known to users, as in the case of the daily-used computer mouse, the remote method can effectively span the space of product properties. The remote method can also benefit from engaging a larger population as, being less resource intensive, can be easily scalable.

What is clear is the dynamics and in-group collaborations involved in a live session differ from those held in a videoconference. During the live session, participants could choose to individually experience the samples or to create temporary pairs and smaller groups to discuss the experience. Also, "trading" mice was commonly seen as an enjoying and motivating experience. In contrast, the videoconference etiquette restrains participants spontaneous engagement and possible interactions were mostly occurring through the whiteboard activities. While splitting the main group into smaller temporary groups by the researchers could be possible, this was not within the design of the experience and would not replicate the natural dynamics of having participants choosing by themselves whom to interact with. Product images were provided by the researchers but, maybe, having the participants bringing their own images or discussing their own mice could have been beneficial and somewhat mimicking the "trading" seen in live sessions.

5.2.3 Panel background effects

The expertise level of the panels played a crucial role in the generation of specific words. The naive panel at Liu consisted of experienced users of computer mice in an office environment, all of whom were academics but lacked expertise in study-related topics. In contrast, the expert panel, composed of professionals in mouse engineering, produced a higher count of words linked to specific and technical aspects of mouse clicking feedback, such as force, tactility, and intensity. Meanwhile, the non-expert panel generated words at a higher level of user experience (UX), focusing on aspects like ergonomics, responsiveness, and distinctiveness. Notably, the expert panel also contributed significantly to high-level UX descriptors, including feelings of engagement, precision, and satisfaction during mouse clicking.

While the expert panel has generated twice as many words per person than the naïve consumer panel, the different settings may have been a major factor. The live setting, with its self-motivational dynamics, may have led to an easier and more immediate word generation than the more constrained videoconferencing settings. Finally, while an expert group may have a larger or more promptly ready verbatim than a naïve consumer panel, it does not necessarily represent a user consumer perspective.

5.2.4 Language and cultural considerations

While some loss of meaning is expected during translation—specifically from Swedish to English—no major or identifiable cultural effects were observed in this study. Although some bias may still be present, with one panel emphasizing ergonomic or distinctiveness aspects while the other focused more on performance, the core meaning associated with the product remained consistent across populations. All participants successfully completed the assessment by following the procedure provided, and despite variations in completion time, all users finished within the expected timeframe. Participants demonstrated an understanding of the procedure outlined in the video tutorial and were able to familiarize themselves with the setup, context, and samples. Importantly, all participants evaluated the eight mouse samples without any missing data points.

5.2.5 User proficiency and familiarity

All participants reported at least an intermediate level of computer proficiency, with over 75% identifying as advanced or expert users. This indicates that the device used in this study was familiar to participants, and the required applications and tasks were well within their capabilities.

5.2.6 Findings on the remote method

The remote method employed in this study demonstrated effective time management and calendarization, necessitating specific IT tools, hardware, and software from the user side. Utilizing Remote Kansei Engineering, the data gathering method was supported by the KESo platform, allowing for a tutored yet self-assessed evaluation process.

5.2.7 Efficiency and scalability

Compared to traditional methods, such as capturing responses through verbose interviews, the remote method proved to be efficient and scalable. Although Kansei Engineering methods require extensive initial work to develop the final survey, the study's design allows for effective scalability. The number of participants at the final evaluation stage, while generating direct data from evaluations, does not significantly impact the researchers' data analysis process. The remote method, characterized by a tutored video and guided on-screen evaluation, does not impose additional time demands on researchers, unlike traditional one-to-one interviews. Users engage in self-assessment without requiring direct involvement from researchers.

A notable exception must be made, namely in providing samples to be evaluated for users. While in a traditional UX setting, the same set of samples (e.g. 8 mice) can be made available and shared among many users, the same cannot be said in a remote design. Here, samples must be afforded to users either by rotating them in a time-consuming process or by replicating them in a resource intensive process. In the latter case, care must be taken for understanding sample consistency. However, in the industry, it is common to manufacture large quantities of experimental samples for UX purposes prior to product market release.

Alternatively, new remote methods could be explored. In a following step to this research case, a more efficient method could have been employed by providing solely the audible experience to the users in a digital platform. Here, the users would perform the UX with their own device, while hearing the desirable experience through headsets (effectively masking their own mouse sound). This method promises far greater scalability and very efficient use of resources.

5.2.8 Potential for repeated measures

The method is replicable within the same user group at different stages of their user journey and experience with the product. It is anticipated that the affective descriptors will remain consistent throughout this journey, although user responses may vary over time. This standardized form of data collection is applicable for future studies involving similar products and contexts, providing a robust framework for ongoing research in user experience and product interaction.

6. Discussion

This study introduces a novel methodology for remotely gathering UX-related data, necessitated by the 2020-2022 pandemic. Although initially designed as a conventional study, the shift to online tools resulted in some incomplete datasets. Nonetheless, the researchers believe the conclusions remain valid and provided further insights on designing UX for the industry.

A limitation of this study is that participants needed prior familiarity with the evaluated products. It is unclear whether this approach would effectively capture data for novel experiences or technologies. The materials used, specifically computer mice, serve as a general case study due to their widespread use. Findings may be extrapolated to other familiar products, such as toothbrushes or power drills, but not to unfamiliar items like VR headsets. However, for expert populations, the methodologies could still apply to specialized products, such as digital sketching tablets.

While remote data collection streamlines the process, it introduces risks, such as uncertainty about whether users are accurately rating the intended samples. Monitoring user actions is limited, necessitating alternative data quality controls. This study employed statistical methods to minimize individual errors and identify outliers, supplemented by live interviews and online tutorials to reduce misunderstandings.

Conversely, online UX data collection eliminates interviewer bias, allowing users to evaluate samples at their own pace in familiar environments, which enhances contextual relevance. However, this approach presents challenges, such as variations in screen resolution, mouse surfaces, and sample layouts, which must be carefully managed.

Remote UX studies require meticulous preparation, as user feedback is limited to predetermined affective channels. This structured survey approach may not capture unique individual insights, and the importance of descriptors may vary among users. While the KE method aims to represent average emotional responses, it risks losing individual nuances by focusing on a generalized set of descriptors derived from initial KE stages.

7. Conclusion

Remote UX data collection can be a viable alternative to traditional methods under suitable circumstances. However, a key challenge remains in establishing the study context, as control over the participant's environment is limited or nonexistent. This shift to individual contexts may result in less focused or immersive experiences compared to live sessions. Nonetheless, a significant advantage is the opportunity to evaluate products in real-world settings, leading to diverse and unique user-driven PUX scenarios. While researchers relinquish some control over the study environment, this may yield valuable UX insights.

To ensure effective remote UX data collection, certain prerequisites must be met: the evaluated products should be common, the affective channels must be broad enough to convey UX remotely, and rating behavior and environments must be effectively controlled. Additionally, outliers and false ratings should be identified and removed from the dataset.

Looking ahead, future studies could explore the remote quantification of UX experiences through biometric data capture, as well as the integration of Virtual or Augmented Reality sessions, provided that supporting devices are available to participants. The methods outlined in this study, particularly the self-assessment virtual guided structured interview, could facilitate the expansion of research to a larger UX panel without local constraints. This approach may also reduce the time and resources typically required for gathering emotional responses compared to conventional UX methods.

For example, researchers envision applying this methodology in a global context, where product samples are shipped to users for remote evaluation. There is also potential to incorporate acoustic cues through virtual or synthesized audio accessible via online platforms, as well as to develop more immersive online presentation and evaluation tools using VR.

The pandemic has highlighted unique scenarios, such as working from home (WFH) arrangements, where multiple individuals may share the same workspace. This context has increased the demand for quieter products, as lower noise emissions from devices like computer mice are now more desirable than in traditional office settings, where background noise can mask such sounds.

Author Contributions

NV and SS: conceptualization of the study and research methods, UX workshops, data gathering, statistical analysis, and manuscript writing. NV: preparation of product samples, experimental measurements for haptics and acoustics. All authors contributed to manuscript revision, read, and approved the submitted version.

Conflict of Interest

The authors declare that they have no conflict of interest.

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To provide a case-study, Logitech (logi) and the Kansei research group at Linköping University (LiU) teamed up with their unique competences. While the research group at Liu possesses a long experience in measuring subjective product experiences with focus on Kansei Engineering methodology, logi possesses moreover a unique knowledge on creating attractive consumer products in the computer peripherals industry. Together it was possible to develop and execute a framework for remote online UX testing.

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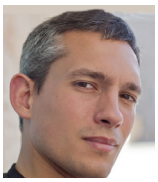
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